Stability and Weight Optimization of a Potential BRICS Currency: An EGARCH-Based Volatility Analysis

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Abstract

This paper examines the financial feasibility of a unified BRICS Currency Unit (UNIT), a proposed composite monetary instrument for Brazil, Russia, India, China, and South Africa. Using an EGARCH(1,1) framework with Student-t distributed errors, applied to daily exchange rate returns from 2017 to 2024, we estimate time-varying volatility, shock persistence, and asymmetry across BRICS currencies against the US dollar. These volatility-based metrics are then used to construct inverse-variance weights, resulting in a basket structure that assigns dominant roles to the Indian Rupee (INR) and Chinese Yuan (CNY).

To complement the short-run financial analysis, we develop a Macroeconomic Vulnerability Index (MVI) using World Bank data from 1999 to 2023. The index captures structural risk across five key areas: inflation, output, fiscal position, energy dependence, and trade exposure. Combining these approaches, we simulate a UNIT index composed of a 60% currency basket and a 40% gold anchor. The resulting index achieves an annualised volatility of 13.62% and a total return of +63.39% over the sample period, indicating meaningful benefits associated with diversification relative to individual BRICS currencies.

Despite these promising results, the findings also highlight important constraints. Structural differences across member economies—most notably the extreme volatility of the Russian ruble following the 2022 sanctions—limit the robustness of a purely volatility-based basket design. The study therefore suggests that while a BRICS currency unit is analytically feasible from a financial modelling perspective, its practical implementation would require careful institutional design and mechanisms to manage geopolitical and macroeconomic asymmetries.

This paper contributes an empirically grounded framework for evaluating composite currency design in emerging market contexts.

Keywords: BRICS Currency Unit, EGARCH(1,1), Exchange Rate Volatility, Inverse-Variance Weighting, Macroeconomic Vulnerability Index, Emerging Markets, Currency Basket Design

1. Introduction

BRICS, comprising Brazil, Russia, India, China, and South Africa, is a group of major emerging economies with increasing influence in global economic and political affairs. The idea of a unified BRICS currency, or at least a structured monetary cooperation mechanism among its member nations, has always remained a serious subject of considerable policy attention since it was informally established in 2009. More recently, geopolitical and macroeconomic developments such as Western sanctions on Russia, ongoing de-dollarisation discussions at BRICS summits, and rising trade in local currencies between these countries have pushed this discussion beyond theory and reinvigorated academic and policy interest in what a workable currency arrangement might look like.

A common currency or shared monetary unit among the BRICS countries presents significant theoretical and empirical challenges. The Optimum Currency Area (OCA) literature,originating with Mundell (1961), outlines key conditions such as factor mobility, trade openness, business cycle synchronisation, and fiscal transfer mechanisms as prerequisites for monetary union. However, the BRICS economies do not meet these requirements. They are geographically dispersed across three continents, exhibit diverse inflation histories, maintain differing fiscal positions, and operate under fundamentally distinct exchange rate regimes.

In this context, a Eurozone style currency arrangement is unlikely to be viable. A more practical alternative is the development of a composite monetary instrument, structured as a weighted basket of member currencies similar to the IMF's Special Drawing Right (SDR).This paper examines whether such an arrangement can be analytically designed and empirically justified in a tractable and policy-relevant manner.

The analysis is carried out along two parallel approaches. First, we estimate EGARCH(1,1) models for each BRICS currency’s daily log returns against the US dollar over the period 2017–2024. The EGARCH framework, introduced by Daniel B Nelson (1991), is particularly appropriate for foreign exchange data, as it captures the asymmetric impact of positive and negative shocks on volatility and does not impose non-negativity constraints on variance. Using these estimates, inverse-variance basket weights are derived to rank currencies based on financial stability.

Second, we construct a Macroeconomic Vulnerability Index (MVI) using World Bank panel data from 1999 to 2023. The index is standardised and aggregated across five structural dimensions: inflation, output, fiscal position, energy import dependence, and trade openness.This macroeconomic layer captures long-term structural context that pure financial volatility models cannot capture.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 presents the methodology, covering both the EGARCH specification and the MVI construction. Section 5 reports empirical results. Section 6 presents the simulated UNIT index. Section 7 discusses policy implications and limitations. Section 8 concludes.

2. Literature Review

2.1 Optimum Currency Areas and Emerging Market Blocs

It is established by the OCA framework that countries benefit from sharing a currency when they have high factor mobility, integrated trade and symmetric responses to shocks. However, if we consider the case of BRICS, these conditions don't apply very well. Velickovski and Stojkov (2011) and Asongu (2014) find limited business cycle synchronization among BRICS countries. However, the design of a basket-based currency unit, as opposed to a full monetary union like the EU, allows us to ignore many of the OCA issues and it would allow the member countries to maintain monetary sovereignty while providing them with a common medium of exchange.

2.2 GARCH-Family Models in Foreign Exchange

Engle's (1982) ARCH paper was extended by Bollerslev (1986) to form the GARCH model. EGARCH was proposed by Nelson (1991) to improve upon the symmetric GARCH variants. This was done by adding a sign term that differentiates negative shocks from positive shocks of equal magnitude. This is much more relevant for currency markets where negative shocks can be much more persistent as compared to positive ones.

A recent cross-country comparison by Jain and Tripathy (2021) demonstrates the better fit of EGARCH over symmetric GARCH models for all five BRICS FX series. Student-t distributed errors are a standard whenever the return series shows excess kurtosis, a common property of high-frequency financial data.

2.3 Composite Currency Baskets

The modern form of the economics of composite currency baskets is based on the SDR (Ainsworth, 1969) and the European Currency Unit (ECU) which was the predecessor of the Euro. Inverse-variance weighting, the allocation of more weight to less volatile inputs is an ideal choice when cross-currency correlation data is absent. This has been recently applied to commodity index design (Erb and Harvey, 2006) and cryptocurrency basket construction (Hileman and Rauchs, 2017). Our application of inverse-variance weights derives from EGARCH conditional variances to a basket of currencies.

2.4 Macroeconomic Vulnerability Indices

Composite vulnerability indices for sovereign risk assessment have a long history, from the IMF's external vulnerability assessments to the World Bank's Country Policy and Institutional Assessments (CPIA). Frankel and Saravelos (2012) applied methods similar to the approach of z-score standardising of indicators within a peer group and aggregating across structural poles, for the early-warning indicators of currency rises. Our construction of MVI compares across BRICS member countries and is designed to complement the financial volatility ranking.

3. Key Contributions

The existing literature on BRICS monetary cooperation has largely approached the question from either a theoretical angle, assessing OCA conditions and finding them unmet, or from a geopolitical one, treating de-dollarisation as a policy ambition without examining what a workable financial instrument might actually look like. This paper attempts to bridge that gap.

Rather than asking whether a BRICS currency is desirable in principle, we ask whether one can be analytically designed in a defensible way. The EGARCH(1,1) framework, applied to each BRICS currency's daily returns from 2017 to 2024, provides a basis for comparing financial stability across member economies that accommodates the asymmetric volatility and excess kurtosis common in high-frequency emerging market data, and which symmetric GARCH models are not well suited to handle.

We also introduce a Macroeconomic Vulnerability Index as a structural complement to the volatility analysis. Most composite currency design studies operate at market data frequency alone. By incorporating a cross-country vulnerability assessment built from World Bank panel data, we are able to distinguish between currencies that are volatile due to transient shocks and those whose instability reflects deeper structural fragility, a distinction with direct implications for basket design and governance.

Finally, the paper treats the Russian Ruble's post-2022 behaviour not as an inconvenient outlier, but as a substantive finding. Its effective exclusion from the UNIT basket, a result of the model rather than an assumption fed into it, illustrates a tension at the heart of the BRICS de-dollarisation project: the same geopolitical conditions motivating the push away from the dollar also introduce the kind of instability that undermines a volatility-weighted basket.

4. Methodology

A two-layered empirical framework is adopted combining high frequency exchange rate data with macroeconomic vulnerability assessment to evaluate the financial feasibility of a BRICS currency basket.

4.1 EGARCH(1,1) Specification

ARCH

Engle (1982) introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model, which established that the variance of financial time series is not constant but depends on past squared residuals. An ARCH(q) process specifies the conditional variance as $\sigma^2 t = \omega + \alpha_1 \varepsilon^2 t-1 + ... + \alpha q \varepsilon^2 t-q$, where large past shocks drive higher current variance. This captures volatility clustering, a pattern that is well documented in financial markets. The limitation of the ARCH model, however, is a practical one: fitting high-frequency data well typically requires a large number of lags, making the model difficult to estimate reliably.

GARCH

Bollerslev (1986) addressed this by adding lagged conditional variance terms to the equation, producing the Generalised ARCH (GARCH) model: $\sigma^2 t = \omega + \Sigma \alpha i \varepsilon^2 t-i + \Sigma \beta j \sigma^2 t-j$. A GARCH(1,1) specification, with just one lag of each, is sufficient for most financial series and keeps the parameter count manageable. The sum $\alpha + \beta$ captures how long the effect of a shock persists, and in financial data this typically falls just below one, reflecting slow but eventual mean reversion in volatility. However, the standard GARCH model treats positive and negative shocks of the same size as having identical effects on variance. In currency markets, this is not a reasonable assumption. Depreciations tend to raise volatility considerably more than equivalent appreciations, and a model that cannot distinguish between the two will systematically misestimate conditional variance.

EGARCH

Nelson (1991) proposed the Exponential GARCH (EGARCH) model to address both the symmetry problem and a secondary issue in standard GARCH, which is the requirement that all parameters be non-negative to prevent the conditional variance from turning negative. By modelling the logarithm of the conditional variance rather than its level, EGARCH ensures that variance remains positive regardless of the sign of the estimated parameters. The EGARCH(1,1) variance equation is:

$$ln(\sigma^2 t) = \omega + \alpha(|zt-1| - E|zt-1|) + \gamma zt-1 + \beta\, ln(\sigma^2 t-1)$$

The return process uses a constant mean model: $rt = \mu + \varepsilon t$, where $\varepsilon t = \sigma tzt$ and zt follows a Student-t distribution with ν degrees of freedom. In the variance equation, ω is the intercept and sets the long-run log-variance level. α captures the magnitude effect of shocks, regardless of their direction. γ is the asymmetry parameter: a negative value means that negative shocks increase variance by more than positive shocks of the same size. β measures how persistent the effect of a shock is over time. The term E|zt−1| is a demeaning correction that ensures α reflects pure magnitude effects without any directional bias.

Why EGARCH?

The choice of EGARCH is motivated by three features of the data and the research context. First, the descriptive statistics in Table 1 show significant excess kurtosis across all five BRICS return series (most notably the Russian Ruble at 781.4), which requires the use of Student-t distributed errors rather than a normal distribution assumption. Second, BRICS currency markets are prone to sharp, asymmetric shocks, including sanctions, capital flight, and political crises, that cause far larger depreciations than comparable appreciations. The γ parameter is necessary to capture this. Third, since inverse-variance basket weights are the primary output of this study, the accuracy of the conditional variance estimates directly determines the quality of the results, and EGARCH is better suited to this than symmetric alternatives in the presence of the structural breaks and one-sided shocks present in the data. Jain and Tripathy (2021) provide direct empirical support for this, showing that EGARCH consistently outperforms symmetric GARCH variants across all five BRICS exchange rate series.

Advantages and Limitations of EGARCH

The main advantages of EGARCH are that the log-variance formulation removes the non-negativity constraints of standard GARCH, the γ term explicitly models the leverage effect, and the Student-t distribution handles fat tails without changing the estimation approach. The limitations are also worth noting. As a univariate model, EGARCH does not account for co-movements between currencies, which is a simplification that is acceptable for inverse-variance weighting but would not hold in a full portfolio optimisation. Post-fit Ljung-Box tests (Section 5.2) also show residual autocorrelation for the Brazilian Real and Chinese Yuan, suggesting the EGARCH(1,1) specification may not be sufficient for those series. Higher-order or regime-switching models could improve this. The model is also sensitive to structural breaks: the 2022 sanctions on Russia produced an estimated persistence of $\alpha + \beta = 2.262$ and an annualised conditional volatility exceeding 24,000%, which reflects the severity of that event rather than an error in estimation, and is why the Russian Ruble receives a near-zero weight in the basket.

4.2 Volatility Metrics and Stability Ranking

From each fitted EGARCH(1,1) model, we extract four stability metrics per currency:

Metric	Formula / Description	Interpretation
Mean Conditional Variance	$E[\sigma^2_t]$ over sample period	Primary ranking criterion: lower = more stable
Unconditional Variance	$\exp(\omega / (1 - \beta))$	Theoretical steady-state variance
Persistence	$\alpha + \beta$	Duration of shock impact; >1 implies explosive process
Half-life (days)	$\ln(0.5) / \ln(\beta)$	Days for shock to decay by 50%

Table 2: Volatility metrics extracted from EGARCH(1,1) models for stability ranking.

Once the EGARCH models are estimated, we extract a set of key metrics to evaluate and compare the stability of each currency.

These include the **mean conditional variance**, which serves as the primary indicator of volatility; the **unconditional variance**, reflecting the long-run level of variability; the **persistence term** (α+β\alpha + \betaα+β), which captures how long the impact of shocks tends to last; and the **half-life of shocks**, which measures the time required for volatility to reduce by half after a disturbance.

Using these metrics, currencies are ranked based on their mean conditional variance, with lower values indicating higher stability. This ranking then forms the foundation for constructing the currency basket in the next stage.

4.3 BRICS Currency Unit (UNIT) Index Simulation

To construct the currency basket, we use an inverse-variance weighting approach, where more stable currencies are given higher weights. In simple terms, currencies with lower volatility are assigned a larger share in the basket.

Assuming that correlations between currencies are negligible, the weights are calculated as:

wi=1/σi2∑j(1/σj2)w_i=\frac{1/\sigma_i^2}{\sum_j (1/\sigma_j^2)}wi=∑j(1/σj2)1/σi2

This method naturally favours currencies with lower estimated variance, ensuring that they play a more dominant role in the overall composition of the basket. As a result, the final allocation reflects a stability-focused structure, where weights follow the same order as the volatility ranking.

4.4 Construction of the BRICS Currency Unit (UNIT)

A synthetic BRICS Currency Unit (UNIT) is constructed as a composite index comprising two components: a currency basket (60%), weighted according to the inverse-variance scheme, and a gold component (40%), incorporated as a stabilising anchor.

All underlying series are normalised to a common base period to ensure comparability. The UNIT index is then computed as a weighted linear combination of these normalised components. The inclusion of gold is motivated by its established role as a store of value and its relatively low correlation with currency returns, which enhances diversification benefits and contributes to overall volatility reduction.

4.5 Macroeconomic Vulnerability Index (MVI)

To complement the short run volatility analysis, a Macroeconomic Vulnerability Index (MVI) is constructed using annual World Bank data covering the period 1999 to 2023. The index is built through a four step process. Following the methodology of Lepers and Sánchez Serrano (2017), who adapt the framework of Aikman et al. (2015) to emerging economies, we constructed a vulnerability index by grouping indicators into four poles:

(i) Valuation pressures and risk appetite: Inflation
(ii) Non-financial sector imbalances: GDP
(iii) Financial sector vulnerabilities: Debt-to-GDP
(iv) External/global vulnerabilities: Net -Trade(Exports-Imports)
Each indicator is standardised using its country-specific means and standard deviations.

Next, for the indicators where a higher value means lower vulnerability, the Z-scores have been sign-flipped so as to make sure that a higher Z-score indicates a higher vulnerability.

Now all of the accumulated Z-scores are converted into a percentile based index ranging from 0 to 1, where higher values indicate greater macroeconomic vulnerability. Each pole score consists of the average of the percentile scores of the indices contained in said pole. (Here only one indicator per pole has been used)

Now the Vulnerability Index has been calculated by averaging across the four poles, providing equal weight to each pole.

5. Data

5.1 Foreign Exchange Data

The daily exchange rates between each of the five BRICS currencies (Brazilian Real (BRL), Russian Ruble (RUB), Indian Rupee (INR), Chinese Yuan (CNY), and South African Rand (ZAR)) and the US dollar have been sourced from Yahoo Finance using the yfinance API for the period between 2017-01-01 and 2024-12-31. The COMEX Gold Futures (GC=F) data for the same time span was used as the underlying commodity for the UNIT index simulation.

Log returns are calculated using the formula $r_t = 100\times\ln(P_t/P_{t-1})$, which measures returns in percentage terms. Table 1 presents descriptive statistics for each return series

Currency	Mean (%)	Std (%)	Skewness	Ex. Kurtosis	JB p-value	ARCH Effect
BRL	0.030	1.038	0.204	4.185	< 0.001	YES ***
RUB	0.026	9.073	0.143	781.4	< 0.001	YES ***
INR	0.011	0.381	0.450	6.100	< 0.001	YES ***
CNY	0.002	0.330	-0.250	7.388	< 0.001	YES ***
ZAR	0.015	1.128	-0.006	98.73	< 0.001	YES ***

*Table 1: Descriptive statistics and pre-fit diagnostics for BRICS FX log returns (2017–2024). Jarque-Bera (JB) tests normality; ARCH-LM (10 lags) tests for conditional heteroskedasticity. *** denotes significance at the 1% level.*

The return series for all five currencies does not satisfy the normality assumption (JB $p < 0.001$) primarily due to excessive kurtosis, particularly in the case of the Russian Ruble (excess kurtosis = 781.4), which can be attributed to the fact that the series is characterized by many tail events due to the economic sanctions imposed on Russia in 2022. The results from the ARCH-LM test confirm significant ARCH effect in all five time series at $p < 0.001$, thereby lending strong empirical backing to the model specification within the GARCH family models

5.2 Macroeconomic Data

Macroeconomic panel data at the country level during the period 1999-2023 was sourced from the World Bank World Development Indicators (WDI). Six variables were collected for each of the BRICS countries, namely annual inflation (consumer price index), GDP (current US$), general government gross debt as % of GDP, merchandise exports (current US$), merchandise imports (current US$), and net international trade (Net exports - Net imports). Four of these variables(Annual inflation, GDP, net international trade (Net exports - Net imports), general government gross debt as % of GDP) were then combined to form a Macroeconomic Vulnerability Index (MVI), as per Section 3.5.

6. Empirical Results

6.1 EGARCH Parameter Estimates

The below table shows the important EGARCH(1,1) parameter estimates, model fit statistics and post-fit test results for all BRICS currency.

Currency	Mean Ann. Vol (%)	Persistence (α+β)	Half-life (days)	Uncond. Var	LB(z_t) p	LB(z^2_t) p	Rank
INR	6.31%	1.191	90.1	0.476	< 0.001	0.195	1
CNY	6.90%	1.356	18.9	2.449	< 0.001	0.025	2
ZAR	15.79%	0.983	2.2	0.909	0.107	1.000	3
BRL	16.42%	1.083	59.6	1.567	< 0.001	< 0.001	4
RUB	24,431%*	2.262	3.5	541.0	1.000	1.000	5

*Table 3: EGARCH(1,1) model results. LB(z_t) and LB(z^2_t) report Ljung-Box test p-values (10 lags) for standardised residuals and squared standardised residuals. *RUB annualised conditional volatility is inflated by the February 2022 structural break.*

As seen above the INR and CNY are the most stable currencies in BRICS by a substantial margin. Their mean annualised conditional volatilities are 6.31% and 6.90% respectively. This is comparable to a second-tier advanced economy currency. On the other hand ZAR and BRL show much higher volatility of about 15-16% annualised which is consistent with their history of exposure to commodity price cycles and domestic political uncertainty.

The RUB presents an extreme case. The EGARCH model absorbs the February 2022 sanctions shock, which caused the ruble to lose approximately 40% of its value in a few days, followed by some recovery under capital controls, as an explosive persistent variance process, with $\alpha + \beta = 2.262$ and an effective annualised conditional volatility exceeding 24.000%. This figure is not an estimation error but a representation of the structural break's impact on the model's conditional variance dynamics. Accordingly, the RUB's inverse-variance weight in the UNIT basket is effectively zero (3.1×10^{-6} percent).

6.2 Diagnostic Assessment

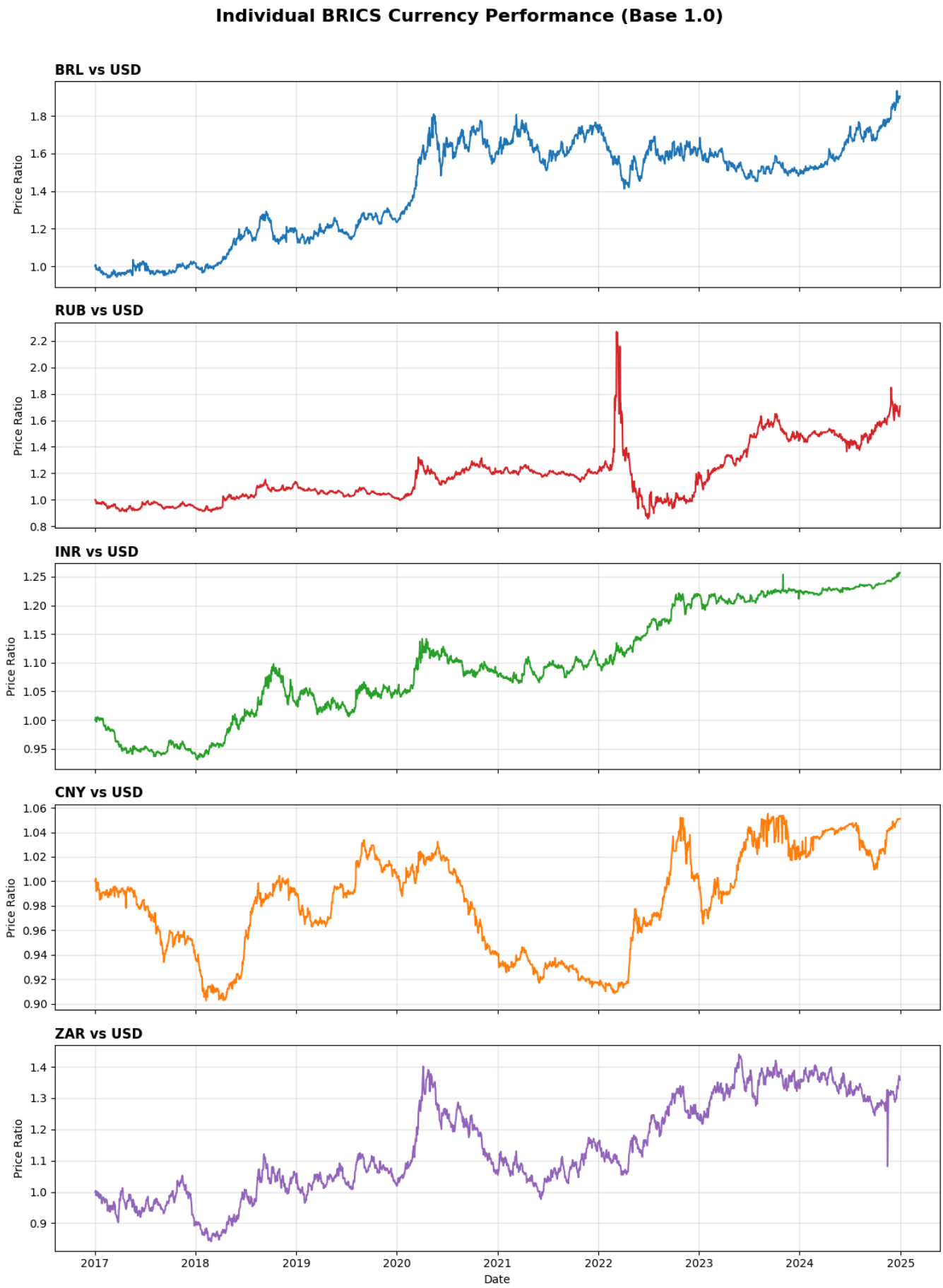

Individual BRICS Currency Performance (Base 1.0)
BRL vs USD
RUB vs USD
INR vs USD
CNY vs USD
ZAR vs USD
Price Ratio
Date
2017
2018
2019
2020
2021
2022
2023
2024
2025

*Figure 1: Annualised conditional volatility for all five BRICS currencies, 2017–2024 (EGARCH(1,1) estimates).***(Graph generated by authors)**

Figure 1above illustrates the time-varying conditional volatility for all five currencies. Three events are very notable: the COVID-19 shock (March–April 2020) elevated volatility across all currencies at the same time, confirming a common global factor; the February 2022 Russia-Ukraine conflict triggered an explosive and sustained surge in RUB volatility; and the ZAR exhibits periodic spikes tied to South Africa's domestic political crises.

Post-fit Ljung-Box diagnostics show a different picture. ZAR standardised residuals pass both the z_t and z^2_t tests ($p = 0.107$ and 1.000 respectively), suggesting good model fit. INR z^2_t also passes ($p = 0.195$). However, BRL and CNY exhibit residual autocorrelation patterns that suggest EGARCH(1,1) may be under-parameterised for these series. This can be improved by using higher-order EGARCH models or regime-switching models.

6.3 Per-Currency Diagnostic Grids

Figure 2 and Figure 3 below present per-currency 2x2 diagnostic grids containing the return series, conditional volatility path, ACF of standardised residuals (z_t), and ACF of squared standardised residuals (z^2_t).

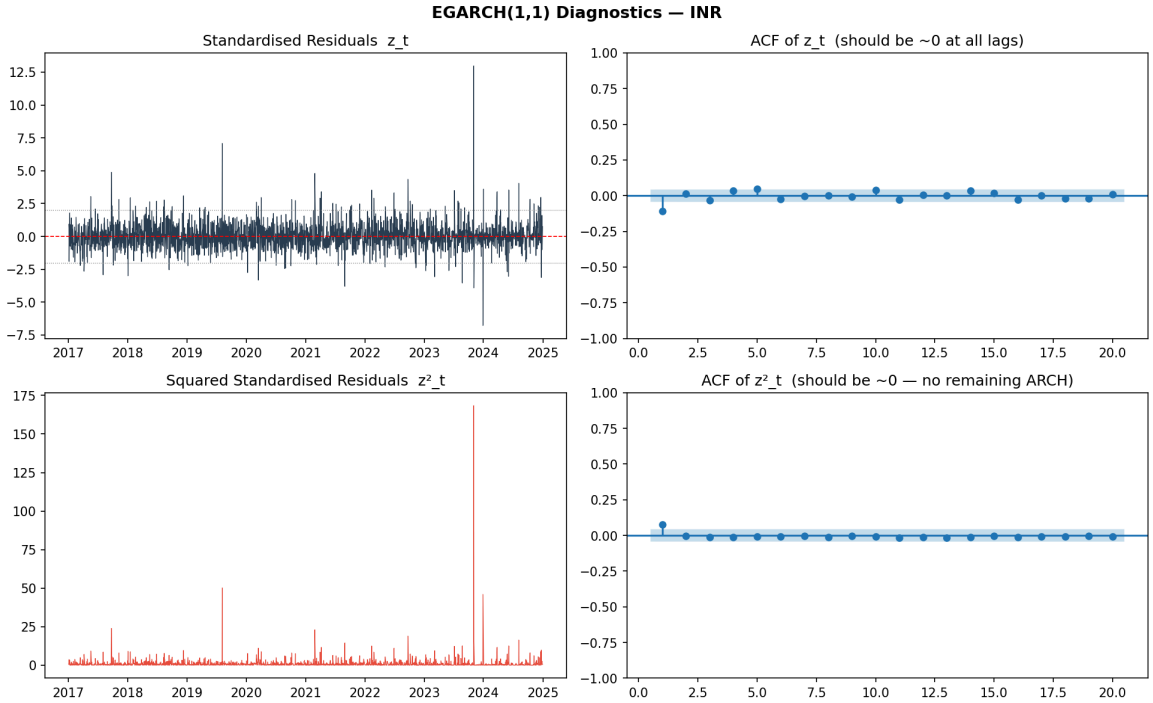


Figure 2: Diagnostic grid for INR — return series, conditional volatility, ACF(z_t), ACF(z^2_t).

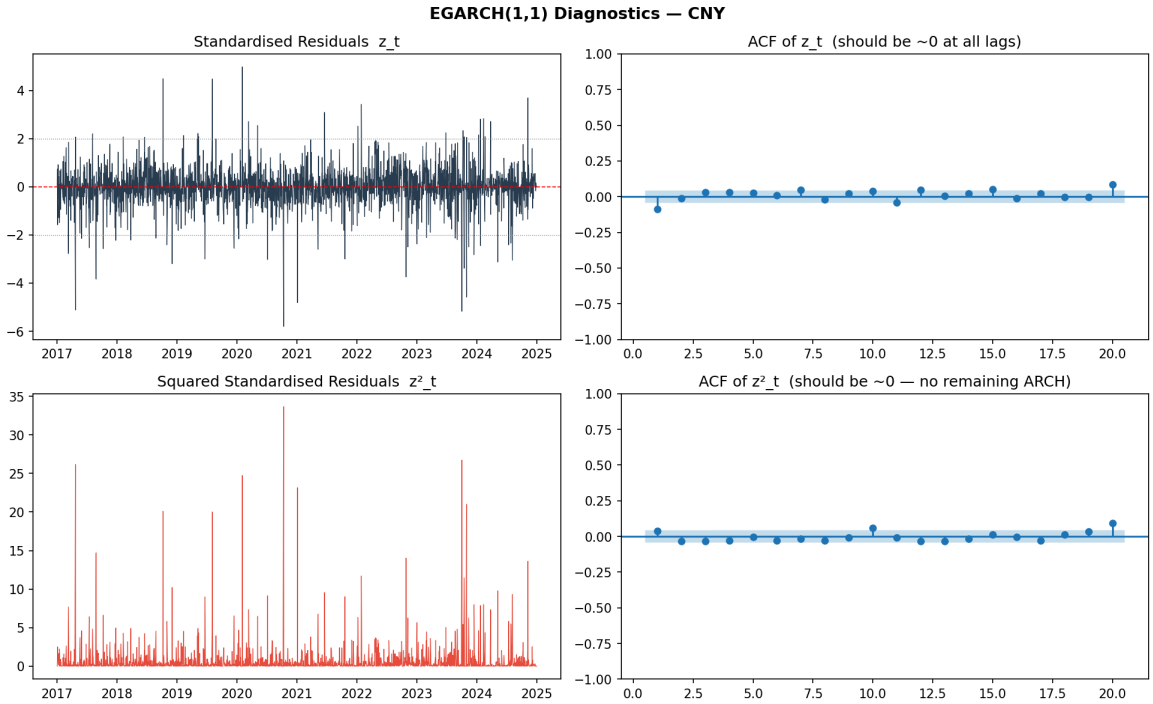


*Figure 3: Diagnostic grid for CNY — return series, conditional volatility, ACF(z_t), ACF(z²_t).***(Graph generated by authors)**

6.4 Stability Ranking and Inverse-Variance Weights

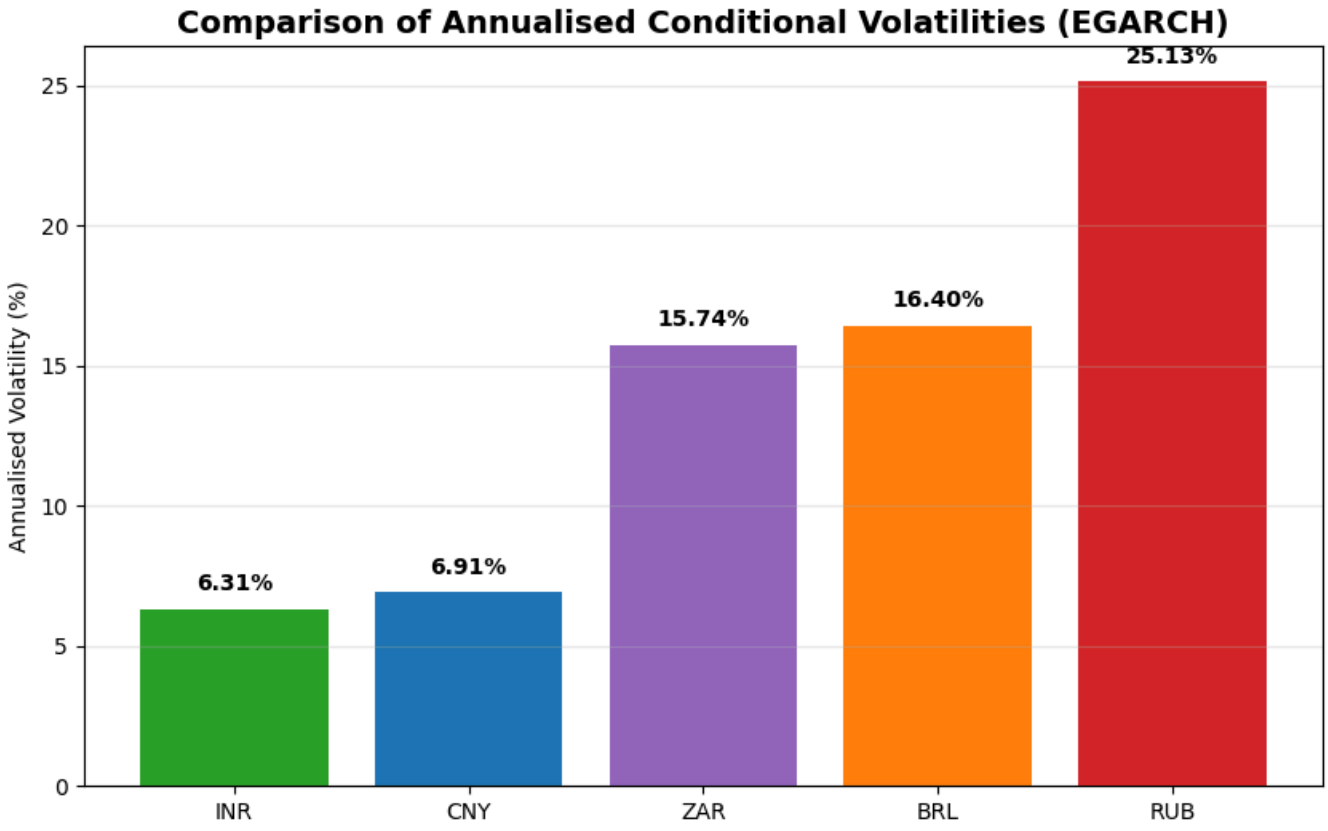


Figure 4: Mean conditional annualised volatility per currency (bar chart) **(Graph generated by authors)**

Figure 4 and Table 4 present the final stability ranking and EGARCH-derived IV weights. The INR and CNY dominate the basket, together accounting for 85.67% of the total weight effectively making them the backbone of

the UNIT. The ZAR and BRL are contributing marginally, 7.45% and 6.88% respectively. The ruble as discussed above is barely present.

Rank	Currency	Mean Cond. Var (daily $\%^2$)	Mean Ann. Vol (%)	IV Weight (%)	
1	INR (Indian Rupee)	0.158041	6.310818%	45.35%	★ Core
2	CNY (Chinese Yuan)	0.189682	6.913753%	37.79%	★ Core
3	ZAR (South African Rand)	0.983496	15.742967%	7.29%	
4	BRL (Brazilian Real)	1.067021	16.397842%	6.72%	
5	RUB (Russian Ruble)	2.505856	25.129179%	2.86%	

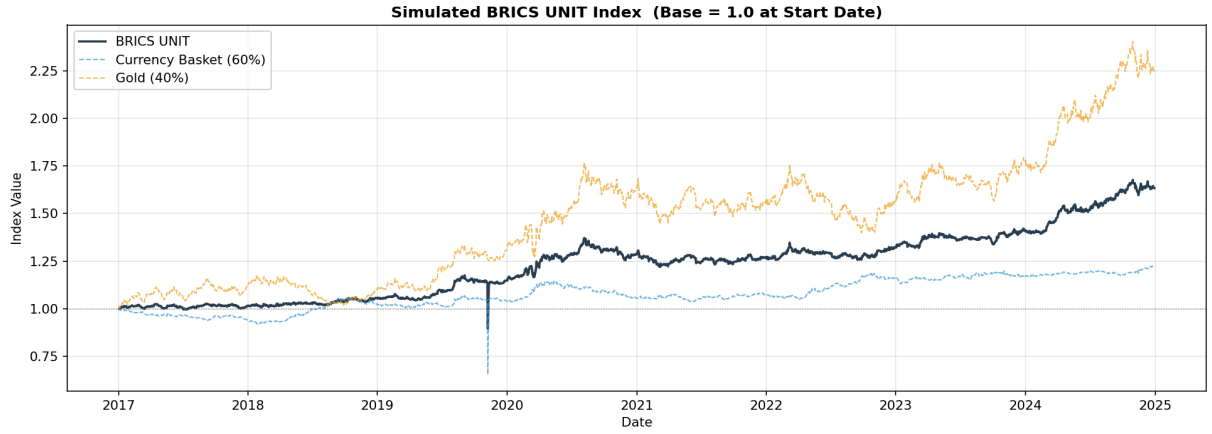
*Table 4: EGARCH stability ranking and inverse-variance basket weights for the BRICS Currency Unit. *RUB figure reflects post-2022 structural break.*

7. The BRICS Currency Unit (UNIT) Index

The BRICS Currency Unit (UNIT) is constructed as a weighted composite of the five currency components, with weights derived from the EGARCH-based stability ranking (accounting for 60% of the index), and a 40% allocation to gold. All components are normalised to a base value of 1.0 in January 2017. Table 5 summarises the key performance metrics of the UNIT over the 2017–2024 simulation period.

Metric	Value
Annualised Volatility of UNIT Index	13.62%
Total Return (2017–2024)	+63.39%
Currency Basket Weight	60% (IV-weighted per EGARCH ranking)
Gold Anchor Weight	40% (COMEX Gold Futures, GC=F)
INR Weight within basket	46.66%
CNY Weight within basket	39.01%
ZAR Weight within basket	7.45%
BRL Weight within basket	6.88%
RUB Weight within basket	≈ 0.00%

Table 5: Simulated BRICS Currency Unit (UNIT) index performance metrics, 201



7–2024.

*Figure 5: Simulated UNIT index (normalised to 1.0 in 2017) versus individual currency components and gold. The gold anchor provides significant upside during the 2020 and 2022 risk-off episodes.***(Graph generated by authors)**

The simulated UNIT index delivers a total return of approximately **+63.39%** over this period, with an annualised volatility of **13.62%**. A significant portion of this performance is driven by gold, which appreciated by roughly 120% over the same timeframe and acted as a stabilising counterweight to the depreciation pressures faced by several BRICS currencies against the US dollar. Within the currency basket, weights are heavily concentrated in the Indian Rupee (46.66%) and Chinese Yuan (39.01%), with smaller contributions from the South African Rand (7.45%), Brazilian Real (6.88%), and an effectively negligible weight assigned to the Russian Ruble.

From a risk perspective, the UNIT demonstrates meaningful diversification benefits. Its volatility compares favourably with that of individual BRICS currencies—particularly the Brazilian Real and South African

Rand—indicating that the basket construction successfully reduces exposure to idiosyncratic currency risk. Even when compared to the more stable components, the UNIT maintains competitive risk characteristics while offering improved return performance.

The behaviour of the index during periods of stress further highlights the robustness of the design. In particular, the February 2022 shock provides a clear illustration: while the Russian Ruble experienced a sharp collapse, the combined stabilising influence of gold and the relatively stable INR–CNY core prevented significant disruption to the overall index. This underscores a key advantage of the basket approach—namely, that extreme movements in any single currency are absorbed rather than transmitted across the system.

A comparison with individual currency components reveals an additional insight. The UNIT not only outperforms the more volatile BRICS currencies such as the Ruble, Rand, and Real in total return terms, but also maintains lower volatility than all but the Indian rupee. From an investment standpoint, this suggests that the UNIT represents a more efficient exposure to BRICS currencies, offering a favourable balance between return and risk relative to holding individual currencies in isolation.

8. Macroeconomic Vulnerability Analysis

While the EGARCH-based volatility ranking captures short-run financial market dynamics, it does not fully account for the underlying structural conditions that shape long-term currency stability. To address this, we construct a **Macroeconomic Vulnerability Index (MVI)** for each BRICS economy using World Bank panel data spanning 1999–2023. This complementary framework enables a more comprehensive assessment by distinguishing between transient financial volatility and persistent macroeconomic fragility.

8.1 Vulnerability Profile Summary

The MVI reveals a more nuanced picture than the volatility-based ranking alone. Although Russia ranks as the most unstable currency in the EGARCH framework—primarily due to the post-2022 sanctions shock—its pre-sanctions macroeconomic fundamentals, including relatively low public debt and sustained energy export surpluses, placed it in the mid-range of BRICS vulnerability. This contrast highlights the importance of distinguishing between **structural vulnerability** and **event-driven financial instability**.

India exhibits the most favourable overall profile when both dimensions are considered. It ranks highest in terms of financial stability and demonstrates a consistent improvement in macroeconomic resilience over the 2010–2023 period. This trend is supported by inflation-targeting reforms, a gradual decline in energy import intensity relative to GDP, and increasing export diversification.

China similarly maintains low vulnerability across several macroeconomic indicators, particularly in inflation and fiscal balance. However, it exhibits relatively high energy import dependence, representing a structural risk. This risk is partially mitigated by China's persistent trade surplus and strong external position.

In contrast, Brazil and South Africa display worsening vulnerability trends over time. Brazil's fiscal position has deteriorated significantly since 2015, with public debt exceeding 90% of GDP by 2023. South Africa faces a combination of structural challenges, including high energy import costs, persistent current account deficits, and subdued economic growth. These macroeconomic weaknesses are consistent with—and help explain—the higher conditional volatilities observed for the Brazilian Real and South African Rand in the EGARCH analysis.

8.2 Structural Implications for UNIT Design

The integration of EGARCH-based volatility metrics with the MVI framework enables a two-dimensional evaluation of currency suitability within the proposed UNIT basket. Ideally, a currency should exhibit both **low short-run volatility** and **low long-run structural vulnerability** to warrant a significant weight in the basket.

Applying this dual-criteria approach to the BRICS sample, the Indian rupee emerges as the only currency that satisfies both conditions consistently over the study period. Other currencies either exhibit elevated financial volatility, structural macroeconomic weaknesses, or both. This reinforces the observed concentration in the EGARCH-derived weights and provides an economic rationale for such outcomes.

From a policy perspective, this framework has important implications for UNIT governance. Rather than relying on static weight assignments, the basket should be subject to periodic review—potentially on an annual basis—incorporating both rolling EGARCH estimates and updated MVI scores. Such a mechanism would allow the system to respond dynamically to changes in both financial and macroeconomic conditions.

In particular, sharp deteriorations in structural stability or sudden increases in financial volatility—such as those observed in the case of the Russian Ruble following the 2022 sanctions—should trigger automatic rebalancing. Embedding such rules within the design of the UNIT would enhance its resilience by reducing discretionary decision-making and improving its ability to adapt to evolving economic and geopolitical conditions.

9. Discussion and Policy Implications

9.1 The Case for and Against a BRICS Currency Unit

While our empirical study suggests cautious optimism for the analytical design of a BRICS currency Unit, it also highlights significant implementation challenges. Despite of the UNIT index performing well as a constructed financial instrument, i.e, diversifying individual currency risk, benefiting from the stability of gold considering its safe-haven properties, and also maintaining a respectable annualised volatility of 13.62% even during periods of major global shocks such as COVID-19 and the Russia-Ukraine war, several structural challenges are evident.
Firstly, the EGARCH-derived basket consisting mainly of INR and CNY (85.67% combined) implies a strong dominance of bilateral currency arrangement between India and China, with other BRICS members as marginal contributors. In the light of ongoing India-China border disputes, this concentration particularly creates geopolitical tensions which any governance framework for the UNIT shall have to address explicitly.
Secondly, the de facto exclusion of the Russian Ruble from the basket following the 2022 sanctions shock illustrates a fundamental vulnerability in the BRICS system. Specifically, the geopolitical factors driving efforts toward de-dollarisation also introduce instability into a volatility-based basket design. Any similar sanctions affecting a major BRICS member would require emergency rebalancing of the UNIT, questioning the institutional capacity to manage such episodes.

9.2 Comparison with the IMF Special Drawing Right

The UNIT is conceptually similar to the IMF's Special Drawing Right (SDR), a basketweighted unit of account comprising the US dollar, euro, Chinese yuan, Japanese yen, and British pound. The SDR's weights are reviewed every five years on the basis of a fundamental approach that complements, without replicating, the volatility-based approach employed here using trade and foreign reserve metrics. A governance framework for the UNIT could usefully incorporate elements of SDR methodology, such as periodic weight reviews, international institutional oversight and clearly defined rules for rebalancing.

9.3 Limitations

The analysis relies on a univariate EGARCH(1,1) model for each currency, which does not account for co-movements between currencies. BRICS exchange rates are influenced by common global factors, and ignoring these interdependencies limits the accuracy of the volatility.

The chosen model specification may not fully capture the underlying dynamics of all currencies. The EGARCH(1,1) model may be insufficient in certain cases, and that approaches—such as higher-order specifications or regime-switching models could potentially provide a better fit to the data.

The results are sensitive to extreme events, such as the 2022 sanctions shock affecting the Russian ruble. Although the EGARCH framework captures this through elevated volatility, such structural breaks can distort long-run estimates and influence the resulting basket weights.

The inverse-variance weighting approach is based solely on volatility, and therefore does not incorporate economic or institutional factors. In practice, considerations such as trade linkages, geopolitical factors, and reserve currency roles also play an important role in determining currency weights.

Macroeconomic Vulnerability Index is constructed using annual data and equal weighting across indicators. This oversimplifies the relative importance of different macroeconomic factors and does not capture higher-frequency changes in economic conditions.

9.4 Limitations and Future Scope

While the EGARCH model gives us a pretty solid mathematical baseline for the UNIT, we have to acknowledge a few structural limitations in our simulation. To make this framework actually viable in the real world, future research needs to tackle a few key areas.

The biggest issue right now is geopolitical. Because our model purely optimizes for mathematical stability, it completely zeroes out the Russian Ruble due to its historical volatility. But realistically speaking, BRICS is a consensus-driven bloc. It is highly unlikely that Russia would endorse a currency where they have zero weight. Future studies really need to look at constrained optimization—like forcing a minimum 5% floor for all founding members. It would be interesting to measure the exact "stability penalty" the index takes just to keep things politically fair. Also, the bloc just expanded into "BRICS+", which changes the whole macroeconomic picture. Bringing in currencies from new members like Saudi Arabia and the UAE—especially since they are pegged to the US Dollar—could completely change the UNIT's volatility profile and maybe even fix the index's current over-reliance on the Rupee and the Yuan.

Then there is the issue of market friction. Our 2017–2024 simulation assumes that rebalancing the basket is a seamless process. However, considering the strict capital controls enforced by both the RBI and the PBOC, the free flow of capital required for this is incredibly complex in practice. To get a more realistic test of how the basket survives in actual forex markets, future models have to factor in liquidity constraints, bid-ask spreads, and transaction costs.

We also need to rethink the anchor. The 40% gold allocation does a great job of stabilizing the simulated index, but scaling that up to handle actual cross-border trade volumes would put a massive strain on sovereign reserves. A logical next step would be testing the UNIT against a broader commodity basket—maybe mixing in crude oil or rare earth metals—so the currency backing actually reflects the export strengths of the expanded bloc.

Lastly, we need to make sure this model isn't just curve-fitted to the relatively stable 2017–2024 window. The only way to really validate its long-term durability is through rigorous out-of-sample stress testing. Running the EGARCH rankings and the resulting basket through actual historical crashes, like the 2008 Global Financial Crisis or the 1997 Asian Financial Crisis, will be the ultimate test of the UNIT's macroeconomic resilience.

10. Conclusion

In this paper, two frameworks have been presented for ascertaining the viability of a BRICS Currency Unit. We have estimated time-varying conditional volatility, leverage effects and shock persistence for each of the BRICS currencies using EGARCH(1,1) models applied to daily FX data from 2017 to 2024. The results place the INR (46.66%) and CNY (39.01%) as the dominant basket anchors, whereas the Russian Ruble's post 2022 variance explosion renders it excluded from any financially coherent basket design.

The simulation of the BRICS Currency Unit index, consisting of a 60% EGARCH-weighted currency basket and a 40% gold anchor, achieves an annual volatility of 13.62% and a total return of 63.69% over the sample period. These statistics demonstrate meaningful risk diversification as compared to individual BRICS currencies and is reasonably competitive with emerging market benchmarks.

Through the complementary Macroeconomic Vulnerability Index, we find that financial volatility and structural fragility do not always move together, Russia had stronger macroeconomic fundamentals before 2022 as compared to post sanction volatility profile, whereas Brazil and South Africa show compounding structural vulnerabilities reflected in their higher EGARCH volatilities. In the index, India emerges as the lone country displaying a favourable profile in both short term financial stability and long term structural resilience dimensions.

The findings of our work suggest that a BRICS Currency Unit is analytically constructible but institutionally challenging. The basket weight being concentrated highly in just two currencies (namely India and China), the geopolitical tensions among BRICS countries and the previously seen capability of sanctions events to destroy the financial stability of individual basket components, all pose serious implementation challenges and risks. A governance framework incorporating timely weight reviews, automatic rebalancing triggers and formal institutional oversight would be essential for any implementation of the concept.

In the future, research should extend in the following three directions:

(i) multivariate GARCH models capturing cross-currency correlations;
(ii) empirical optimisation of the currency-gold weight split;
(iii) incorporation high-frequency macroeconomic data into a dynamic MVI that can be updated in near real-time alongside EGARCH estimates.

These extensions would move the BRICS currency unit concept from a research instrument toward an actual implementable credible policy proposal.

References

Ainsworth, G. (1969). The International Monetary Fund and the Special Drawing Right. Finance & Development, 6(2), 18–24.

Asongu, S. A. (2014). Are proposed African monetary unions optimal currency areas? Real, monetary and fiscal policy convergence analysis. African Journal of Economic and Management Studies, 5(3), 253–286.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics, 31(3), 307–327.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica, 50(4), 987–1007.

Erb, C. B., & Harvey, C. R. (2006). The strategic and tactical value of commodity futures. Financial Analysts Journal, 62(2), 69–97.

Frankel, J. A., & Saravelos, G. (2012). Can leading indicators assess country vulnerability? Evidence from the 2008–09 global financial crisis. Journal of International Economics, 87(2), 216–231.

Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. Journal of Finance, 48(5), 1779–1801.

Goudarzi, H., & Ramanarayanan, C. S. (2011). Empirical analysis of the impact of the global financial crisis on the Indian stock market volatility with GARCH models. International Journal of Economics and Finance, 3(2), 13–19.

Hileman, G., & Rauchs, M. (2017). Global cryptocurrency benchmarking study. Cambridge Centre for Alternative Finance.

Jain, A., & Tripathy, T. (2021). Estimating and forecasting BRICS currency exchange rate volatility. Cogent Economics & Finance, 9(1), 1985777.

Kenen, P. B. (1969). The theory of optimum currency areas: An eclectic view. In R. A. Mundell & A. K. Swoboda (Eds.), Monetary Problems of the International Economy (pp. 41–60). University of Chicago Press.

McKinnon, R. I. (1963). Optimum currency areas. American Economic Review, 53(4), 717–725.

Mundell, R. A. (1961). A theory of optimum currency areas. American Economic Review, 51(4), 657–665.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. Econometrica, 59(2), 347–370.

Velickovski, I., & Stojkov, A. (2011). Feasibility of a common currency in the Western Balkans. International Journal of Finance and Economics, 16(5), 481–496.

World Bank (2024). World Development Indicators. The World Bank Group. https://databank.worldbank.org/source/world-development-indicators